EVPN-VXLAN Q&A – Arrcus Prep

Sriram Dendukuri · 8 October 2026 · 25 questions, tuned to your Aruba AOS-CX experience

How to use: cover the answer, say yours out loud (control plane → data plane → failure → verify), then compare. ★ = areas where past Arrcus candidates failed: dataplane, asymmetric IRB, MAC move, DF election, "what is each route type for". Orange boxes link the answer to your AOS-CX work.

1. Fundamentals

Q1. Why use EVPN instead of flood-and-learn VXLAN?

- Flood-and-learn: remote VTEPs configured statically or via multicast; MACs learned by flooding in the data plane – lots of BUM, no MAC mobility control, no ARP suppression.
- EVPN: MP-BGP control plane advertises MACs/IPs (type 2) and discovers VTEPs (type 3) → less flooding, ARP suppression, MAC mobility, multitenancy (L3VNI/VRF), multihoming, policy with RT/route-maps.

> **AOS-CX:** static VXLAN (vtep-peer) exists only on 5420/6200; static VTEPs are not supported when EVPN is enabled.

Q2. Describe the VXLAN encapsulation. What overhead and MTU do you need?

- Outer Ethernet (14) + outer IPv4 (20) + UDP (8, dst **4789**) + VXLAN header (8: I-flag + 24-bit **VNI**) = **50 bytes** (54 with an outer 802.1Q tag; 70 with IPv6 outer).
- 24-bit VNI → ~16 million segments vs 4094 VLANs.
- UDP **source port** = hash of the inner flow → ECMP entropy across the spine.
- Underlay MTU ≥ inner MTU + 50; normally set 9100–9216 on fabric links.

Q3. Define VTEP, L2VNI, L3VNI, EVI / MAC-VRF.

Term	Meaning
VTEP	Encap/decap point, identified by a loopback IP (source ip of interface vxlan)
L2VNI	Bridge domain; maps 1:1 to a VLAN on the leaf
L3VNI	One per tenant VRF; carries routed traffic in symmetric IRB
EVI / MAC-VRF	The EVPN instance for a VLAN/VNI: its own RD, RT, MAC table

Q4. Describe a typical underlay/overlay design.

- **Underlay:** OSPF/IS-IS or eBGP (unique ASN per leaf) – only needs loopback reachability between VTEPs, with ECMP.
- **Overlay:** MP-BGP L2VPN EVPN (AFI 25 / SAFI 70), usually iBGP between loopbacks with spines as **route reflectors**, or eBGP overlay.
- Must have send-community extended (RT, router-MAC, MAC mobility are extended communities).

> **AOS-CX:** router bgp → address-family l2vpn evpn → neighbor x activate / send-community extended; update-source loopback.

2. Route types ★

Q5 ★. List the route types and what each is used for.

Type	Name	Used for
1	Ethernet A-D (per ES / per EVI)	Multihoming: mass withdraw, aliasing, split horizon label
2	MAC/IP advertisement	MAC learning; MAC+IP for ARP suppression and host routes (symmetric IRB)

Type	Name	Used for
3	Inclusive Multicast (IMET)	VTEP discovery + BUM flood list per VNI (ingress replication)
4	Ethernet Segment	ES peer discovery + DF election
5	IP Prefix	Subnets / external prefixes into the L3VNI; silent hosts; DCI

> **AOS-CX:** AOS-CX natively uses **types 2, 3 and 5**. Dual-homing is VSX; type-1/4 interop only on 8325/10000.

Q6. What is inside a type-2 route?

- RD, ESI (0 for single-homed / VSX), Ethernet Tag, **MAC**, optional **IP**, label 1 = **L2VNI**, label 2 = **L3VNI** (only when IP present and symmetric IRB).
- Extended communities: **RT** (import), **Encapsulation = VXLAN**, **Router-MAC** (for symmetric IRB), **MAC Mobility** (sequence number, sticky flag).
- MAC-only route → bridging; MAC+IP → ARP suppression table + /32 host route in the VRF.

Q7 ★. What does a type-3 route carry and what happens if one is missing?

- Originated per VNI by each VTEP; carries the **PMSI Tunnel attribute** – tunnel type ingress replication, tunnel ID = VTEP IP.
- Each VTEP builds its **head-end replication list** per VNI from received type-3s.
- Missing type-3 from Leaf3 for VNI X → Leaf3 not in others' flood list → **BUM/ARP fails** to hosts on Leaf3 in that VNI, while known unicast (via type-2) may still work.

> **AOS-CX:** show evpn vtep-neighbor, show interface vxlan vteps (flood list), show bgp l2vpn evpn.

Q8. Type-5 vs type-2 with IP – when do you need type-5?

- Type-2 MAC+IP = a **host** (/32) tied to a MAC.
- Type-5 = an **IP prefix** independent of MACs: subnets, external routes from a border leaf/firewall, summarisation, **silent hosts** (the subnet still attracts traffic so the owning leaf can ARP).
- Type-5 carries the L3VNI and router-MAC; next-hop = advertising VTEP.

> **AOS-CX:** redistribute host-route / local-svi; border VTEP leaking EVPN routes to an external VRF.

Q9 ★. RD and RT in EVPN. What breaks with auto-RT and an eBGP underlay/overlay?

- RD makes routes unique per VTEP (usually router-ID:EVI); RT controls import into the MAC-VRF / IP-VRF.
- Auto RT is typically **ASN:VNI**. With eBGP and a **different ASN per leaf**, each leaf derives a different RT → routes not imported.
- Fix: manual RTs, or the vendor option to ignore/rewrite the ASN in auto-RT.
- Any manual RT on one leaf that differs from the rest → that VNI breaks (no type-2/3 import).

> **AOS-CX:** evpn → vlan 10 → rd auto, route-target export/import auto (or manual values).

3. Data plane ★

Q10 ★. Walk a packet: Host A (Leaf1) → Host B (Leaf2), same VLAN 10 / VNI 10010.

1. Host A ARPs for B: Leaf1 answers locally if ARP suppression has B's MAC+IP (type-2); otherwise floods via HER.
2. A sends frame to B's MAC; Leaf1 MAC lookup in VLAN 10 → remote entry learned from type-2 → next-hop VTEP Leaf2, VNI 10010.
3. Leaf1 encapsulates: inner frame unchanged; VXLAN VNI 10010; UDP 4789, src port = hash; outer IP Leaf1 → Leaf2 loopback.
4. Spines route on the outer IP only (ECMP).
5. Leaf2 decaps, VNI 10010 → VLAN 10, MAC lookup → access port → Host B.

> **Say it:** *Leaf1 finds B's MAC in the MAC-VRF from a type-2 pointing to Leaf2, wraps the frame with VNI 10010 in UDP 4789 to Leaf2's loopback, the spines just route the outer header, and Leaf2 maps the VNI back to VLAN 10 and bridges to B.*

Q11. How is BUM traffic handled? What is VXLAN split horizon?

- **Head-end/ingress replication:** ingress VTEP sends one unicast copy to every VTEP in the type-3 list for that VNI.
- **Multicast underlay:** VNI mapped to a group; less replication load, needs PIM in the underlay.
- Split horizon: a VTEP never re-floods BUM received from a VXLAN tunnel back into other tunnels – only to local ports.
- ARP suppression and type-2 learning reduce BUM.

Q12. How does ARP suppression work? What about silent hosts?

- Leaf builds an ARP/ND cache from type-2 MAC+IP routes; local ARP requests for known remote IPs are answered by the leaf, not flooded.
- Unknown target → flood as normal.
- Silent host (never sends): no type-2 → others can't reach it at L3 in symmetric IRB unless the **subnet is advertised as type-5**, so traffic reaches a leaf that then ARPs locally.

> **AOS-CX:** arp-suppression / nd-suppression under evpn vlan; show evpn mac-ip.

4. IRB and gateways ★

Q13 ★. Asymmetric vs symmetric IRB.

	Asymmetric	Symmetric
Ingress VTEP	Routes + bridges; encaps with **destination L2VNI**	Routes into VRF; encaps with **L3VNI**
Egress VTEP	Bridges only	Routes again (L3VNI → VRF → dest VLAN)
Inner DMAC	Destination host MAC	Egress VTEP **router-MAC**
Config	All VLANs/VNIs + SVIs on all leaves	Only local VNIs + one L3VNI per VRF
Return path	Uses the other VNI ("asymmetric")	Same L3VNI both ways
Scale	All MAC/ARP everywhere – poor	Better; supports type-5

Q14 ★. Walk an inter-subnet packet in symmetric IRB: Host A (VLAN 10, Leaf1) → Host C (VLAN 20, Leaf2).

1. A sends to its gateway MAC (anycast gateway on Leaf1).
2. Leaf1 routes in VRF-Red: C's /32 learned from type-2 MAC+IP (or type-5) → next-hop Leaf2, L3VNI 50000, router-MAC of Leaf2.
3. Encapsulate: VNI 50000, inner SMAC = Leaf1 router-MAC, inner DMAC = **Leaf2 router-MAC**, TTL decremented.
4. Leaf2 decaps, L3VNI → VRF-Red, routes to VLAN 20, rewrites to C's MAC (TTL decremented again) and sends.

Q15. What is a distributed anycast gateway and why the same MAC everywhere?

- Every leaf has the same gateway IP **and** MAC for a subnet, so routing happens at the first-hop leaf (no hair-pinning).
- Same MAC means a VM that moves to another leaf keeps a valid ARP entry for its gateway – no re-ARP, no traffic loss.
- Gateway MAC is not advertised as a normal host MAC (or is marked to avoid mobility conflicts).

> **AOS-CX:** active-gateway ip <ip> and active-gateway ip mac <mac> on each SVI.

Q16. "Asymmetric VNI" / VNI mismatch – what happens if Leaf1 maps VLAN 10 to VNI 10010 but Leaf2 maps it to 10011?

- VNIs are globally significant in a standard fabric: the encapsulated VNI must match the receiver's mapping.
- Leaf2 receives VNI 10010 for which it has no bridge domain (or the wrong one) → drop; type-2/3 for the VNI also won't line up → one-way or no traffic.
- (Downstream-assigned VNI exists only in special DCI cases.)

5. MAC mobility ★

Q17 ★. A VM moves from Leaf1 to Leaf2. What happens in the control plane?

- VM sends GARP/RARP; Leaf2 learns the MAC locally.
- Leaf2 advertises type-2 with **MAC Mobility extended community, sequence = previous + 1** (first advertisement has none = 0).
- All VTEPs prefer the **highest sequence**; Leaf1 sees a higher sequence for its local MAC, withdraws its route and flushes.
- Equal sequence from two VTEPs → lowest VTEP IP wins (tie-break).

Q18. What are duplicate-MAC detection and the sticky bit?

- If a MAC moves N times in M seconds (common default **5 moves in 180 s**), it is flagged **duplicate** and frozen – advertisement stops, logs raised; recovery by timer or manual clear.
- Typical cause: two hosts with the same MAC, a loop, or misconfigured multihoming.
- **Sticky/static** flag: a static MAC cannot be moved; competing advertisements are ignored – protection against spoofing.

> **AOS-CX:** mac-move-detection count timer; EVPN MAC dampening. Test: move a host back and forth, check the sequence number increments and the freeze threshold.

6. Multihoming

Q19 ★. How does dual-homing work in your AOS-CX fabric (VSX) vs standards ESI multihoming?

	VSX (AOS-CX)	ESI multihoming (type-1/4)
Seen by fabric as	One **logical VTEP** – shared anycast VTEP IP	Two VTEPs, same ESI
Load-balancing	Underlay ECMP to the shared IP	Aliasing (per-EVI type-1)
BUM toward host	Arrives once at logical VTEP – no DF needed	DF election (type-4)
State sync	ISL syncs MAC/ARP; keepalive for split-brain	Control plane only
Failure	Peer takes over; ISL for orphan ports	Mass withdraw (per-ES type-1)
Scope	2 switches, vendor-specific	Multi-vendor, 2+ PEs

Be honest: "My hands-on dual-homing was VSX; I know the ESI model conceptually." Verify the VSX details against your own test notes.

Q20. Explain DF election, aliasing, mass withdraw and local bias (conceptual).

- **DF election:** PEs on the same ES find each other via type-4; default: order IPs, DF for VLAN V = index (V mod N). DF forwards BUM to the ES; non-DF drops it.
- **Aliasing:** per-EVI type-1 lets remote VTEPs ECMP to all PEs on the ES even if only one learned the MAC.
- **Mass withdraw:** withdraw one per-ES type-1 on link failure → remotes drop that next-hop for all MACs at once.

- **Local bias (RFC 8365):** VXLAN has no ESI label, so BUM from a local ES is always sent to local ES ports by the ingress PE, and a PE drops BUM from a peer VTEP that shares the ES.

Q21. Dual-homed server gets duplicate broadcasts. Causes?

- Both PEs think they are DF: type-4 not exchanged (ES-Import RT mismatch, BGP down), **ESI typo**, different VLAN lists.
- Local bias/split horizon broken – peer VTEP not recognised as sharing the ES.
- In VSX: both members flooding because the logical VTEP isn't shared correctly or the ISL/keepalive state is wrong.

7. Troubleshooting ★

Q22 ★. Two hosts in the same VNI on different leaves cannot ping. Your checklist?

1. **Underlay:** loopback-to-loopback ping between VTEPs; ECMP paths; MTU.
2. **Overlay session:** show bgp l2vpn evpn summary – Established, routes received.
3. **VTEP discovery:** type-3 for that VNI from the remote leaf; remote leaf in flood list.
4. **MAC learning:** local MAC learned on the access port; type-2 sent and received; imported into the MAC-VRF (RT match).
5. **Mapping:** VLAN ↔ VNI identical on both leaves; interface vxlan up; source ip correct.
6. **ARP:** ARP suppression entry present; GARP/ARP seen.
7. Capture on the spine/leaf uplink for VXLAN UDP 4789 with the right VNI.

> **AOS-CX:** show bgp l2vpn evpn summary · show evpn evi detail · show evpn vtep-neighbor · show evpn mac-ip · show interface vxlan vni · show interface vxlan vteps · show mac-address-table.

Q23 ★. Same-VNI traffic works, but inter-VNI (routed) traffic fails. What do you check?

- L3VNI configured and mapped to the same VRF on both leaves (VLAN/VNI for L3VNI with routing).
- Type-2 routes carry label 2 (L3VNI) and the **Router-MAC** community.
- VRF route-targets import/export match.
- Anycast gateway SVI present and up on the ingress leaf.
- Host /32 or type-5 subnet present in the VRF table of the ingress leaf.
- Asymmetric design: destination VLAN/VNI must exist on the ingress leaf.

Q24. Small pings work across the fabric, large transfers hang. Why?

- Underlay MTU too small for the 50-byte VXLAN overhead; DF set on the outer IP → large frames dropped (often silently).
- Test: ping with DF and increasing size from host and between VTEP loopbacks.
- Fix: raise fabric MTU (9100–9216) or lower host MTU.

8. DCI and testing

Q25. How do you extend EVPN between two data centres?

- **L2 DCI:** stretch VNIs via border VTEPs (type-2/3 across sites), control BUM and MAC scale.
- **L3 DCI:** only type-5 prefixes between sites (VRF leaking/route-targets at border), no MAC stretch.
- Considerations: RT design per site, next-hop rewrite at border gateways, loop prevention, failure isolation.

> **AOS-CX:** L2 DCI, L3 DCI with IVRL at border VTEPs, symmetric IRB with external connectivity (border leaks EVPN routes to an external VRF).

If asked "how would you test an EVPN-VXLAN release?"

Area	Test ideas
Control plane	Type-2/3/5 origination and withdrawal, RT import/export (auto and manual), RR

Area	Test ideas
	reflection, route-maps (match vni, set local-pref)
Data plane	Known unicast, BUM via HER, ARP/ND suppression, ECMP hashing, MTU boundary, captures verifying VNI
IRB	Symmetric and asymmetric inter-VNI routing, anycast gateway, router-MAC, TTL, type-5 external routes
Mobility	MAC/IP move, sequence numbers, duplicate detection, sticky MACs
Resiliency	VSX member failure, ISL/keepalive failure (split-brain), uplink/spine failure, BGP session flap – measured loss in ms
Scale	Max VNIs, VRFs, MACs, ARP entries, VTEPs; churn
Interop	Against other vendors' EVPN – route formats, RT, ESI handling